SL. No.	Paper	Work Done	Technology Used	Result	Limitation	Research Gap
1	A Stacking-based Ensemble Framework for Automatic Depression Detection using Audio Signals	Built an audio-based depression model using stacking with multiple classifiers; created new dataset; extracted eGeMAPS features; compared with baseline models.	openSMILE, eGeMAPS, KNN, NB, SVM, DT, Logistic Regression (meta-classifier), SOX preprocessing	Accuracy 79.1%, F1-score 80%; ensemble outperformed single models.	Small dataset; unimodal; controlled reading task only.	Need larger datasets; multimodal extension; deeper stacking approaches.
2	Detection of Major Depressive Disorder Based on a Combination of Voice Features	Collected real clinical voice data; extracted 6552 openSMILE features; selected features using ROC + PCA; built logistic regression classifier.	openSMILE, PCA, ROC-AUC filtering, Logistic Regression	~90% training accuracy; ~80% testing accuracy; stable PC-based model.	Controlled scripted speech; risk of overfitting; sensitive to environment.	Real-world conversational speech needed; robustness improvements; multimodal integration.
3	An Investigation of Emotional Speech in Depression Classification	Studied emotional dimensions (arousal, valence, dominance); combined affect ratings with acoustic features; SVM classification; emotion-based data selection.	AVEC 2014 dataset, affect ratings, LLDs, SVM, MATLAB	Manual arousal ratings → 70% accuracy; adding acoustic features → +5% improvement.	Depends on human emotion raters; small controlled dataset; limited acoustic analysis.	Automatic emotion recognition needed; larger natural speech datasets; deeper affect models.

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4	Detecting Depression Using Vocal, Facial & Semantic Communication Cues	Combined audio, facial actions, and text semantics; extracted formant & MFCC dynamics; used FAUs & GloVe embeddings; fused using Gaussian staircase regression.	AVEC 2016 dataset, Kalman formants, MFCC delta features, FAUs, GloVe embeddings, Gaussian regressor	Dev F1 = 0.81; Test F1 = 0.70; multimodal outperformed unimodal.	Segmentation errors; audio noise variations; computational complexity.	Need joint end-to-end multimodal deep learning models; cleaner datasets; real-time system design.